the relationship known as “mutualism,” both organisms benefit from the relationship.

A tree and its ants

In the mid-1960s, Daniel Janzen, a young American ecologist, became fascinated by the amazing mutualistic relationship between acacia trees and ants in eastern Mexico. His research was one of the first in-depth studies of such an interaction. The two partners were the swollen-thorn acacia and the acacia ant, which lives in the bullhorn-shaped thorns of the tree. He found that queen ants sought out unoccupied shoots, cut a hole in one of the swollen thorns, and laid their eggs, sometimes leaving the thorn to forage on the tree’s nectar. Larvae hatching from the eggs then fed on the acacia’s leaf-tips, with their rich supplies of sugars and proteins. The larvae later metamorphosed into worker ants. In time, all the tree’s thorns became occupied, with up to 30,000 ants living in a colony.

Janzen showed that, unless the acacia ants were present to defend it, the swollen-thorn acacia lost the ability to withstand damage caused by insects that ate its leaves, stems, flowers, and roots. Without the ants, a tree would be stripped of its leaves and die within six months or a year. Because it could not sustain growth, it was also likely to be shaded out by competing trees. Janzen clipped thorns and cut or burned shoots to remove ants from trees, and found that the ants moved back in when new thorns started to grow.

In return for food and shelter, the ants provided two services for the tree: they defended its foliage from leaf-eating insects and ate potentially competitive tree seedlings growing close by. Janzen described the acacias and their ants as “obligate mutualists,” meaning that one species would die out without the other. If the ants were removed, the swollen-thorn acacia would have no means of defending itself. And if the acacia trees were removed, the ants would have no home.